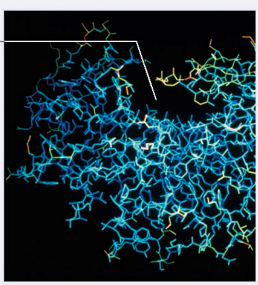
3 IN THE SMALL INTESTINE

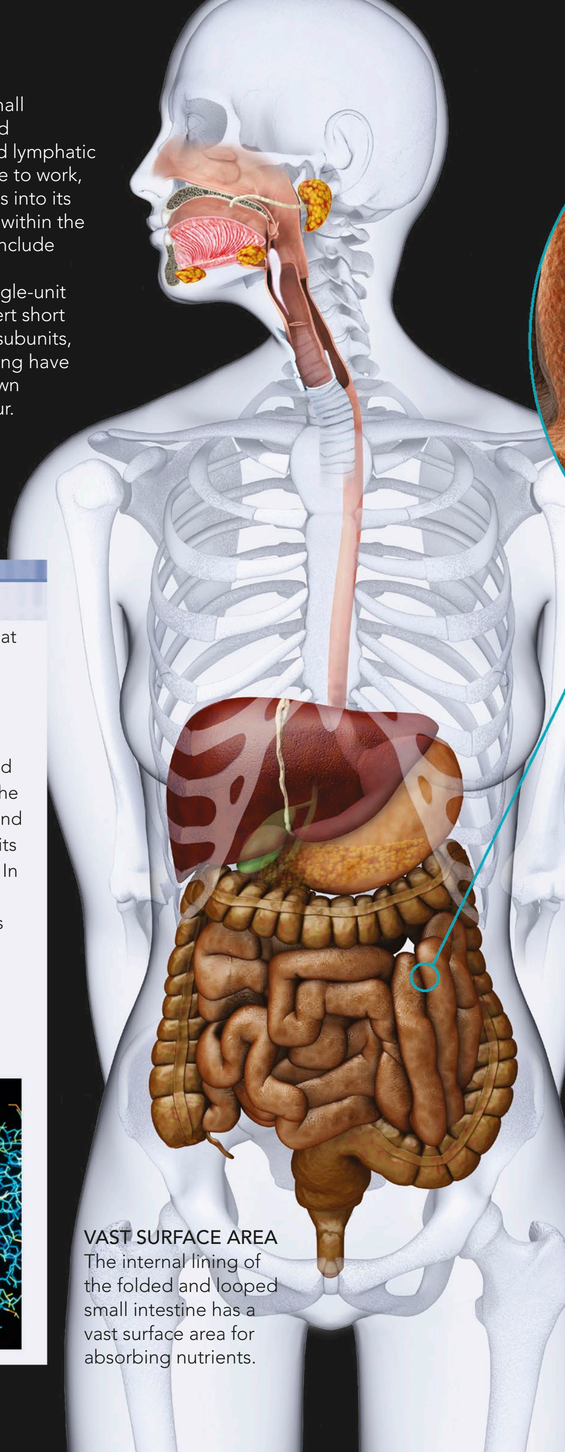
After the duodenum, the remainder of the small intestine is the site for the final breakdown of food substances and their absorption into the blood and lymphatic fluids. The pancreatic juices and bile fluids continue to work, but the small intestine releases few further enzymes into its inner passage, the lumen. Instead, its enzymes act within the lining cells, and on their surfaces. These enzymes include lactase and maltase, which break down the double (disaccharide) sugars, lactose and maltose, into single-unit glucose and galactose. Intestinal peptidases convert short peptide chains (originally from proteins) into their subunits, amino acids. The fingerlike villi of the intestine lining have surface cells bearing smaller projections of their own (microvilli), where some of these final changes occur.

HOW ENZYMES WORK

An enzyme is a biological catalyst – a substance that boosts the rate of a biochemical reaction, but remains unchanged itself. Most enzymes are proteins. They affect the reactions of digestive breakdown, and also the chemical changes that release energy and build new materials for cells and tissues. Each enzyme has a specific shape due to the way its long chains of subunits (amino acids) fold and loop. The substance to be altered (the substrate) fits into a part of the enzyme known as the active site. In the case of digestion, the enzyme may undergo a slight change in 3-D configuration that encourages the substrate to break apart at specific bonds between its atoms.



PEPSIN
A computer model of this digestive enzyme shows the active site as the gap at the top. A protein molecule will slot in here and break apart.



VAST SURFACE AREA
The internal lining of the folded and looped small intestine has a vast surface area for absorbing nutrients.